Heredity

The big idea

Children resemble their parents but are never exact copies. **Heredity** is the passing of traits from parents to offspring; **variation** is the difference between individuals. Gregor Mendel worked out the rules using garden peas.

Mendel's experiments

Mendel chose pea plants with clear contrasting traits: tall/short, round/wrinkled seeds, violet/white flowers.

MONOHYBRID CROSS: TALL × SHORT

Parents: TT (tall) × tt (short). **F_1**: all Tt, all **tall**. Tall is **dominant**, short is **recessive**.

Self-cross F_1 (Tt × Tt): **F_2** genotypes TT : Tt : tt = **1 : 2 : 1**; phenotypes tall : short = **3 : 1**.

The short trait reappears in F_2, so it was hidden, not lost.

DIHYBRID CROSS: ROUND YELLOW × WRINKLED GREEN

F_1: all round yellow. F_2 phenotype ratio **9 : 3 : 3 : 1** (round yellow : round green : wrinkled yellow : wrinkled green).

New combinations appear, so the traits are **inherited independently**.

KEY WORDS

Gene: a section of DNA that codes for a protein, which controls a trait.

Each trait is controlled by **two copies** of a gene, one from each parent. A

dominant trait shows even if only one copy is present.
Genotype: the genes (TT, Tt). **Phenotype**: what you see (tall).

How genes work: a gene for a growth hormone produces the enzyme. An efficient version means more hormone and a taller plant.

Sex determination

In humans, women have two X chromosomes (**XX**) and men have one X and one Y (**XY**). Every egg carries an X. A sperm carries either X or Y. If an X sperm fertilises the egg, the child is a girl (XX); if a Y sperm does, a boy (XY). So **the father's sperm decides the sex of the child**.

In some animals the environment decides: in some reptiles, the incubation temperature of eggs determines the sex of the offspring.

RATIOS ARE PROBABILITIES

A 3 : 1 ratio means the chance of a tall plant is 3/4 for each offspring. It doesn't mean that exactly 3 out of every 4 plants will be tall in a small sample.

Quick recap

- Traits come in pairs of genes, one from each parent.
- Monohybrid F_2: 3 : 1 (phenotype), 1 : 2 : 1 (genotype). Dihybrid F_2: 9 : 3 : 3 : 1.
- XX = female, XY = male; the father's sperm decides.

Practice questions

1. In Mendel's monohybrid cross, the F_2 phenotypic ratio is
   (a) 1 : 2 : 1 (b) 3 : 1 (c) 9 : 3 : 3 : 1 (d) 1 : 1
2. A plant with genotype Tt is
   (a) tall (heterozygous) (b) short (c) pure tall (d) medium height
3. The dihybrid F_2 ratio is
   (a) 3 : 1 (b) 1 : 2 : 1 (c) 9 : 3 : 3 : 1 (d) 1 : 1 : 1 : 1
4. The sex of a human child is determined by
   (a) the mother (b) the chromosome in the father's sperm (c) the environment (d) the number of chromosomes
5. A gene is a segment of
   (a) protein (b) DNA (c) RNA only (d) cytoplasm

Answers: 1. b 2. a 3. c 4. b 5. b